

Predicting Parcel Dwell Time at Parcel Lockers

– Using Statistical and Machine Learning Methods

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Abstract

Parcel locker networks are a growing component of last-mile logistics, yet their effective capacity is constrained not by the number of compartments but by how long parcels remain inside them. This thesis addresses the problem of predicting parcel dwell time; the duration between delivery to a locker and customer pickup; using survival analysis and machine learning methods applied to a dataset of approximately 100,000 parcel-level records.

Five models of increasing complexity were implemented and evaluated: a Kaplan–Meier baseline, a Cox proportional hazards model, Weibull and Log-Normal accelerated failure time (AFT) models, a Random Survival Forest, and an XGBoost-AFT model. Models were compared using the concordance index for discrimination and the integrated Brier score for calibration. Feature importance was assessed through Weibull acceleration factors, permutation-based variable importance from the Random Survival Forest, and SHAP values from XGBoost-AFT.

All covariate models achieved concordance index values in the narrow range 0.649–0.663, substantially above the chance level of 0.5 but converging regardless of model complexity. This pattern reflects a low-signal dataset: the available features; delivery timing, parcel size, and compartment size; explain only a limited portion of the variance in individual pickup times. The remaining variance is attributed to unobserved factors such as the customer's personal schedule and distance from the service point. Despite limited discrimination, the best-performing models produce well-calibrated survival curves, with the XGBoost-AFT model achieving the lowest integrated Brier score (0.152) and highest concordance index (0.663). The Weibull AFT model is preferred on distributional fit grounds, with an AIC advantage of 8,734 over the Log-Normal alternative.

Feature importance results are consistent across all three methods. Delivery day of week is the dominant predictor: Friday deliveries exhibit median dwell times approximately 2.4 times longer than Monday deliveries due to the weekend. Delivery hour and parcel size are the next most important factors, while booking-to-delivery lead time and delivery month carry negligible predictive information. These findings provide a data-driven answer to which operational factors most influence locker occupancy and suggest that targeted interventions; such as escalating notifications for Friday deliveries; could meaningfully reduce long-tail occupancy.



Acknowledgments

I want to express my sincere gratitude to my supervisor, Jose M Pena, for his constant guidance and time throughout the thesis. It was a valuable experience that significantly helped me for applying the knowledge I gained in my master's program and also learning beyond the scope of the program. It would not have been possible without my family, thank you for the continuous support throughout these past months. I will always be grateful to my parents' constant efforts that they have made to ensure that I can pursue my goals and get the best educational experience possible. Also, I want to thank my friends for their support and encouragement.

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1 Introduction

The rapid growth of e-commerce has fundamentally transformed consumer expectations around parcel delivery. Customers now expect fast, flexible, and reliable options for receiving purchases, placing enormous operational pressure on postal and logistics providers. In response, parcel locker networks have emerged as a cost-effective and scalable solution that enables self-service collection without requiring a home delivery attempt. PostNord, the leading postal and logistics company in the Nordic region, operates an extensive network of parcel lockers and service points across Sweden. As parcel volumes grow, efficient utilisation of this fixed infrastructure has become a critical operational challenge.

A key bottleneck in parcel locker operations is the duration a parcel occupies a locker compartment; commonly referred to as *dwelt time*. Defined as the elapsed time between a parcel's delivery and the customer's subsequent pickup, dwell time directly determines how many parcels can be processed through the network over any given period. Prolonged dwell times mean compartments remain occupied for extended periods, reducing effective throughput and increasing the probability that incoming deliveries are rejected due to unavailability of suitable compartments. These rejected deliveries carry direct revenue costs and erode customer satisfaction.

Despite its operational importance, dwell time is rarely modelled explicitly in current locker management systems. Existing assignment strategies are largely rule-based and static: parcels are placed in available compartments on a first-fit basis without any prediction of how long those compartments will be occupied. This reactive approach misses the opportunity to leverage rich historical data; encompassing parcel characteristics, service point attributes, temporal demand patterns, and customer pickup behaviour; to produce data-driven predictions.

1.1 Background

Parcel locker systems consist of a fixed set of compartments at a service point location, classified by size (typically small, medium, large, and slim variants). When a parcel is delivered, it is assigned to a matching compartment, which remains occupied until the customer collects it. The challenge arises because customer pickup behaviour is stochastic and heterogeneous: some customers collect within hours of the delivery notification, while others delay for days. This variability, combined with the fixed compartment supply, creates the capacity mismatch that drives assignment failures.

Dwell time is the central determinant of locker throughput. Empirical evidence from deployed locker systems confirms that shorter dwell times allow compartments to cycle significantly more deliveries per day [1]. Conversely, parcels that remain in lockers beyond three days; which account for approximately 15% of the PostNord dataset; occupy capacity for disproportionately long periods and are the primary driver of assignment failures during busy periods.

Survival analysis, the statistical framework for modelling the time until a well-defined event occurs, provides a natural methodological foundation for dwell time prediction. Originally developed for medical and reliability applications [2, 3], survival analysis has since been applied productively across a range of time-to-event problems in industry, including customer churn prediction [4] and locker pickup behaviour [5]. The framework is particularly well-suited to logistics data because it handles *censored observations* correctly: parcels not yet collected at the time of data extraction contribute partial information about pickup timing rather than being discarded.

PostNord’s operational data provides a rich basis for this analysis. The dataset contains approximately 100,000 parcel-level records, with features including delivery timestamps, compartment and parcel size categories, and observed dwell times for completed pickups.

1.2 Motivation

The practical motivation for this research is straightforward: assignment failures cost money and erode customer trust. Undelivered parcels must be rerouted and redelivered at additional cost, and customers who experience repeated rejections are more likely to switch to alternative providers. A predictive model for dwell time provides a quantitative basis for understanding when compartments are likely to become available, and for identifying which parcel and operational characteristics most strongly drive long occupancy.

There is also a methodological motivation. The Amazon locker system demonstrates what is achievable with machine-learning-based dwell time prediction: combining predicted pickup time distributions with a linear programming optimiser for capacity reservation produced a 9% year-over-year increase in locker throughput during their 2018 holiday season [6]. While the capacity reservation component is outside the scope of this thesis, the predictive modelling stage; producing accurate per-parcel dwell time distributions and identifying their key drivers; is directly relevant.

Furthermore, the problem sits at the intersection of survival analysis, logistics demand forecasting, and feature engineering on structured tabular data, an area that has received limited academic attention in the parcel locker context. Producing well-validated survival models and a principled feature importance analysis contributes both to the applied literature on smart locker systems and to the methodological literature on survival analysis for logistics applications.

1.3 Aim

This research aims to develop and evaluate statistical and machine learning models for predicting the parcel dwell time in lockers, and to demonstrate how these predictions can be used to improve locker utilization and reduce assignment rejections. These models will be evaluated using standard metrics for survival analysis, and the results will be analyzed. The following research question will be addressed:

1.4 Research Questions

1. How can we model the probability distribution of parcel dwell times?



2. What factors; including parcel type, service point location, time of day; influence the dwell time of parcels in lockers?

Chapter 2 presents the theoretical foundations: the survival analysis framework, the family of models employed, and the evaluation metrics. Chapter 3 describes the data, feature engineering decisions, and the implementation of each model in detail sufficient for replication. Chapter 4 presents the experimental results and discusses findings relative to the two research questions. Chapter 5 concludes with a summary of contributions, limitations, and directions for future work.



2 Theory

Understanding why parcels sit in lockers longer than they should requires more than intuition, it requires a principled mathematical framework for modelling how long it takes for an event to happen and which factors speed it up or slow it down. That framework is *survival analysis*, a branch of statistics originally developed for biomedical research [2, 3] that has proven equally applicable in engineering reliability, marketing analytics, and logistics. This chapter develops the theoretical tools used throughout the thesis, starting from first principles and building up to the machine-learning-augmented models that form the methodological core of the work.

The structure mirrors the modelling pipeline. Section 2.1 introduces the survival and hazard functions that describe dwell time mathematically, and explains why censoring parcels that have not yet been collected at the time of data extraction must be handled carefully rather than discarded. Section 2.2 covers the non-parametric and semi-parametric baselines: the Kaplan-Meier estimator for exploratory analysis and the Cox proportional hazards model as the primary reference point, which, notably, has already been applied to the exact problem of parcel locker pickup in recent literature [5]. Section 2.3 develops parametric accelerated failure time (AFT) models, with a focus on the Weibull and Log Normal families whose right-skewed shape matches the distributional structure of the company data. Section 2.4 describes the machine-learning extensions; XGBoost-AFT and Random Survival Forests; that capture the non-linear interactions among parcel, and temporal features. Finally, Section 2.5 defines the evaluation metrics used to compare the models: the concordance index, the Brier score and its integrated form, integrated Brier score.

2.1 Foundations of Survival Analysis

Before describing any particular model, it is worth asking: why use survival analysis at all, rather than ordinary regression? The straight-forward answer is censoring. In the dataset, although parcels are collected by the time the data snapshot was taken but the data points which needs to be predicted are assumed to be inside the locker and having an unknown pickup time. This shows that the prediction will happen on censored data. For these observations, we know that the pickup time exceeded some threshold, but we do not know the actual value. This type is well-known in wide range of applied problems: patient time of death or disease onset in medical field, time to component failure in reliability engineering, and time to customer churn in marketing analytics [4]. In all these cases, as in parcel locker operations,

ignoring censored observations or treating them as observed events produces biased estimates of the underlying time-to-event distribution. Survival analysis helps in correctly predicting this through the likelihood contributions described below. A similar argument applies in Amazon’s locker capacity management system, where the modelling challenge is explicitly framed around uncertain pickup time windows that are only partially observed at any decision point [6].

2.1.1 The Survival Function

Let $T \geq 0$ be a random variable representing the time from parcel delivery to customer pickup. The *survival function* $S(t)$ gives the probability that a parcel is still in the locker at time t :

$$S(t) = P(T > t), \quad t \geq 0. \quad (2.1)$$

By definition $S(0) = 1$ and $S(t)$ decreases monotonically toward zero. Reading this curve from left to right tells an intuitive operational story: the steeper the drop in the first few hours after delivery, the more customers collect their parcels quickly; a long, flat tail indicates a population of slow collectors who tie up compartments for extended periods. This distinction matters enormously for throughput. Empirical evidence from locker deployments in the United States suggests that *shorter dwell times allow compartments to be cycled more frequently*, directly increasing the number of deliveries a locker can accept per day [1].

2.1.2 The Hazard Function

While $S(t)$ describes the global picture, the *hazard function* $h(t)$ captures the instantaneous risk of pickup at time t conditional on the parcel still being in the locker:

$$h(t) = \lim_{\Delta t \rightarrow 0} \frac{P(t \leq T < t + \Delta t \mid T \geq t)}{\Delta t} = \frac{f(t)}{S(t)}, \quad (2.2)$$

where $f(t) = -S'(t)$ is the density of T . The hazard is not a probability; it is defined as a rate, and can take any non-negative value. Its shape carries substantive information. A high hazard shortly after delivery, decreasing over time, would suggest that customers who do not collect promptly are increasingly unlikely to do so, this is the expected pattern for parcel pickup. In survival analysis terms, this corresponds to a *decreasing* hazard function.

The survival and hazard functions are related through the cumulative hazard $H(t) = \int_0^t h(u) du$:

$$S(t) = \exp(-H(t)). \quad (2.3)$$

Understanding how covariates modulate $h(t)$, whether a parcel delivered on a Friday has a lower pickup rate over the weekend than one delivered on a Monday, for instance, is exactly what RQ2 of this thesis is asking.

2.1.3 Handling Censored Observations

A parcel observation (t_i, δ_i) consists of a time t_i and an event indicator $\delta_i \in \{0, 1\}$, where $\delta_i = 1$ if the pickup was observed and $\delta_i = 0$ if the observation is *right-censored* (the parcel was still in the locker at time t_i). The correct likelihood contribution of such an observation is:

$$\mathcal{L} = \prod_{i=1}^n [f(t_i)]^{\delta_i} [S(t_i)]^{1-\delta_i}. \quad (2.4)$$

Observed pickups contribute their density $f(t_i)$; censored parcels contribute only the probability of surviving to t_i . All models in this thesis are estimated by maximising a version of this likelihood, ensuring that partial information from uncollected parcels is used rather than discarded.

2.2 Non-parametric and Semi-parametric Baseline Models

Any modelling study benefits from robust baselines; simpler models that set a floor on what constitutes genuine predictive improvement. In survival analysis, the natural baselines are the Kaplan-Meier estimator for descriptive analysis and the Cox proportional hazards model for covariate-adjusted inference. Importantly, the Cox model has already been applied in the parcel locker setting: [5] developed a Cox Proportional Hazard Regression model to analyse how a reward-and-punishment incentive scheme affects customer pickup behaviour, treating the survival of a parcel in a locker as the outcome of interest. This gives direct methodological precedent for the approach taken in this research, and the results of that study showing that incentive schemes can accelerate pickup among some customer segments but not others also illustrate the kind of heterogeneity across customer groups that motivates the richer, covariate-aware models of later sections.

2.2.1 Kaplan-Meier Estimator

The Kaplan-Meier (KM) estimator [3] is the standard non-parametric maximum likelihood estimator of the survival function $S(t)$. Let $t_{(1)} < t_{(2)} < \dots < t_{(k)}$ denote the distinct observed event times, and let d_j and n_j denote the number of pickups and the number of parcels still in a locker just before time $t_{(j)}$. The KM estimate is:

$$\hat{S}(t) = \prod_{j: t_{(j)} \leq t} \left(1 - \frac{d_j}{n_j}\right). \quad (2.5)$$

This makes no distributional assumptions and is consistent under non-informative censoring. Its role in this research is exploratory: stratifying the KM curve by parcel size, or day-of-week delivery and other categorical covariates immediately reveals whether these groups have meaningfully different pickup patterns. For instance, if small and large parcels have visually distinct KM curves. This gives us an idea on whether the covariates have some signal that answers part of RQ2 before any model is fitted. The limitation of the KM estimator is that it cannot incorporate multiple continuous covariates simultaneously or generate predictions for new observations, which is why it serves as an exploration tool rather than a deployed predictor. The log-rank test [7] is used to formally assess whether stratified curves differ significantly.

2.2.2 Cox Proportional Hazards Model

The Cox proportional hazards (PH) model [2] is the primary model of applied survival analysis. It models the hazard for a parcel with covariate vector $\mathbf{x} \in \mathbb{R}^p$ as:

$$h(t | \mathbf{x}) = h_0(t) \exp(\boldsymbol{\beta}^\top \mathbf{x}), \quad (2.6)$$

where $h_0(t)$ is an unspecified baseline hazard and $\boldsymbol{\beta}$ is the vector of log-hazard-ratio coefficients. The baseline hazard $h_0(t)$ is left completely unspecified, which is defined as the model semi-parametric and avoids any distributional assumption about the shape of the dwell time. The key structural assumption is that covariates multiply the baseline hazard by a time-constant factor; the *proportional hazards assumption*:

$$\frac{h(t | \mathbf{x}_i)}{h(t | \mathbf{x}_j)} = \exp(\boldsymbol{\beta}^\top (\mathbf{x}_i - \mathbf{x}_j)) = \text{const.} \quad (2.7)$$

This assumption is verified empirically. The standard way to do so is using *Schoenfeld residual test* [8]: if the residuals are uncorrelated with time, the PH assumption holds; if they show a systematic pattern, it is violated. Estimation proceeds via the *partial likelihood* [9], which conditions on the observed ordering of pickup events:

$$\mathcal{L}_{\text{partial}}(\boldsymbol{\beta}) = \prod_{i: \delta_i=1} \frac{\exp(\boldsymbol{\beta}^\top \mathbf{x}_i)}{\sum_{j \in \mathcal{R}(t_i)} \exp(\boldsymbol{\beta}^\top \mathbf{x}_j)}, \quad (2.8)$$

where $\mathcal{R}(t_i)$ is the set of parcels still in the locker just before time t_i . The Breslow estimator subsequently recovers the baseline cumulative hazard [10], enabling individual survival curve predictions.

Furthermore, the ℓ_2 and ℓ_1 regularization penalties were tuned through grid search hyperparameter tuning to determine the best values. More details can be found in Section 3.3.2 where the grid is shown.

In the applied context of parcel lockers, [5] found that the Cox model was effective at identifying which customer segments respond to pickup incentives with elderly customers purchasing daily necessities showing the strongest response and younger non-regular workers showing the weakest. This kind of group-level insight is exactly what the hazard-ratio output of the Cox model supports, and it motivates checking the proportional hazards assumption carefully when applying the model to the company data, since different customer segments may exhibit qualitatively different time-varying pickup dynamics. This literature directly motivates the choice of Cox model as an interpretable baseline for second research question.

2.3 Parametric Accelerated Failure Time Models

The Cox model is powerful but says nothing about the underlying distribution of dwell times. When the data is right-skewed, as parcel pickup times typically are, with a concentration of quick pickups and a long tail of delayed ones, parametric models that directly specify this distribution offer additional benefits: they produce smoother predictions, allow extrapolation beyond observed time ranges, and directly parameterise the shape of the dwell time distribution.

Accelerated failure time models approach the problem from a different angle to the Cox model. Rather than modelling how covariates multiply the hazard, they model how covariates stretch or compress the time axis.

2.3.1 The Accelerated Failure Time Framework

An AFT model specifies the log of the event time as a linear function of covariates plus a random error term:

$$\log T = \boldsymbol{\beta}^\top \mathbf{x} + \sigma \varepsilon, \quad (2.9)$$

where $\sigma > 0$ is a scale parameter and ε ; an error term; follows a distribution that determines the parametric family. The corresponding survival function is given by:

$$S(t | \mathbf{x}) = S_0 \left(\frac{t}{\exp(\boldsymbol{\beta}^\top \mathbf{x})} \right), \quad (2.10)$$

where $S_0(\cdot)$ is the baseline survival function. The quantity $\phi(\mathbf{x}) = \exp(\boldsymbol{\beta}^\top \mathbf{x})$ is the *acceleration factor*: values above 1 stretch the time axis (slower pickup of parcels), values below 1 compress it (faster pickup of parcels). This time-scale interpretation is operationally intuitive: if a parcel in a large compartment has an estimated acceleration factor of 1.5 relative to a small one, it is predicted to stay in the locker 50% longer, which directly informs how many large-compartment slots should be reserved at a given service point. Or parcel delivered on a Friday may have an acceleration factor of 1.4 relative to one delivered on a Monday, suggesting that it is likely to stay in the locker 40% longer. This is more intuitive than the hazard ratio from the Cox PH model and it can be used as the primary output for RQ2.

Applied studies in time-to-event modelling for customer retention have shown that Weibull and Log-Normal AFT models generalise well to service time distributions with similar right-skewed shapes. The Gamma AFT model was also considered but later was dropped due to memory constraints during the fitting of the model.

2.3.2 Weibull AFT Model

Setting $\varepsilon \sim \text{Gumbel}(0, 1)$ in Eq. (2.9) yields the Weibull distribution, whose survival and hazard functions are:

$$S(t | \mathbf{x}) = \exp\left(-\left(\frac{t}{\lambda}\right)^k\right), \quad h(t) = \frac{k}{\lambda} \left(\frac{t}{\lambda}\right)^{k-1}, \quad (2.11)$$

with scale $\lambda > 0$ and shape $k > 0$. The shape parameter determines the hazard's behaviour: when $k < 1$ the hazard decreases over time, when $k = 1$ it is constant (exponential), and when $k > 1$ it increases over time. Parcel dwell times are expected to exhibit a decreasing hazard: customers are most likely to collect their parcel in the first day or two after delivery, and the conditional pickup rate falls among those who have not picked up yet. This expectation aligns with the agent-based simulation results of [11], which found that the probability of pickup drops substantially for parcels that remain in lockers beyond a threshold time, creating the right-skewed distribution that motivates the Weibull family.

A further reason to consider the Weibull specifically is that it is the only distribution that simultaneously satisfies both the proportional hazards assumption and the AFT assumption [12]. This means Weibull hazard-ratio estimates can be compared directly against Cox model estimates, providing a useful consistency check.

2.3.3 Log-Normal AFT Model

Setting $\varepsilon \sim \mathcal{N}(0, 1)$ yields the Log-Normal distribution:

$$S(t | \mathbf{x}) = 1 - \Phi\left(\frac{\log t - \mu}{\sigma}\right), \quad (2.12)$$

where $\Phi(\cdot)$ is the standard normal CDF and $\mu = \beta^\top \mathbf{x}$ is the predicted value of $\log T$. Unlike the Weibull, the Log-Normal hazard is non-monotone: it rises to a peak before declining. This additional flexibility makes it a competitive alternative when the strict monotone hazard of the Weibull does not hold. Both models are estimated by maximising Eq. (2.4); model selection is performed via the Akaike Information Criterion [13]. The equation for calculating AIC is given by Eq. (2.13) where k is the number of estimated parameters and $\hat{\ell}$ is the maximised log-likelihood.

$$\text{AIC} = 2k - 2\hat{\ell}. \quad (2.13)$$

2.4 Machine-Learning-Augmented Survival Models

The parametric and semi-parametric models above assume that covariate effects enter the hazard or log-time scale in a log-linear or additive way. In practice, the factors that influence parcel dwell time likely interact in ways that are difficult to specify in advance. Whether a parcel was delivered on a Friday matters more when the following Monday is a public holiday; the effect of compartment size may differ across service points located in residential suburbs versus transport hubs; and customer pickup speed likely depends on a joint combination of notification timing, parcel priority, and day of week rather than on any single factor in isolation. Machine-learning-augmented survival models retain the survival framework and replace the linear predictor with a flexible data adaptive function. They are designed precisely to capture such interactions without requirement to specify them manually.

The Amazon locker case provides direct evidence that ML-augmented approaches outperform simpler heuristics in the locker setting. [6] combined machine learning dwell-time predictions with a linear programming optimiser for capacity reservation, producing an 8% reduction in prediction error and a 9% year-over-year increase in locker throughput during the 2018 holiday season.

2.4.1 XGBoost-AFT Model

The XGBoost-AFT model [14] replaces the linear predictor in Eq. (2.9) with a gradient-boosted tree ensemble $\mathcal{T}(\mathbf{x}) = \sum_{m=1}^M f_m(\mathbf{x})$:

$$\log T = \mathcal{T}(\mathbf{x}) + \sigma Z, \quad (2.14)$$

where Z follows a chosen distributional family (normal, logistic, or extreme value). The ensemble is trained by minimising the negative log-likelihood of the AFT model via second-order gradient boosting with ℓ_1/ℓ_2 regularisation. Survival curves are recovered analytically from the model’s predicted log-time output $\hat{\mu}_i = \mathcal{T}(\mathbf{x}_i)$ and an empirical scale $\hat{\sigma}$. [14] demonstrate on multiple benchmark datasets that XGBoost-AFT achieves competitive or superior performance compared to linear AFT baselines, with particular gains on datasets where covariate effects are non-linear.

2.4.2 Random Survival Forests

Random Survival Forests (RSF), introduced by [15], extend the random forest framework to right-censored survival data by replacing the standard splitting criterion with the *log-rank test statistic*, which measures how well a candidate split separates the survival curves of the two resulting child nodes. Concretely, at each internal node, the algorithm searches over all candidate features and split-points s ; for each candidate it partitions the node’s observations into a left child and a right child, and evaluates the log-rank statistic:

$$L(s) = \frac{\left(\sum_{j=1}^k (d_{j,L} - e_{j,L})\right)^2}{\sum_{j=1}^k v_j}, \quad (2.15)$$

where the sum runs over all k distinct event times $t_{(1)} < \dots < t_{(k)}$ observed in that node, $d_{j,L}$ is the number of observed events (pickups) in the left child at time $t_{(j)}$, and $e_{j,L}$ is the corresponding expected count under the null hypothesis that the two children share the same underlying survival curve. The denominator $\sum_{j=1}^k v_j$ accumulates the null-hypothesis variance of the log-rank numerator across all event times, so that $L(s)$ is a standardised measure; a large value of $L(s)$ indicates that the candidate split s creates two child nodes with substantially different survival curves, and the split that maximises $L(s)$ is chosen. Each tree is grown on a bootstrap sample drawn with replacement from the training data, with $\sqrt{p}$ features randomly selected as candidates at each node, where p is the total number of covariates.

Within each leaf node, the cumulative hazard function is estimated using the Nelson–Aalen estimator applied to the bootstrap observations that fall into that node. The ensemble prediction for a new observation $\mathbf{x}$ then averages these per-leaf estimates across all B trees:

$$\hat{H}(t | \mathbf{x}) = \frac{1}{B} \sum_{b=1}^B \hat{H}^{(b)}(t | \mathbf{x} \in \ell^{(b)}(\mathbf{x})), \quad (2.16)$$

where $\ell^{(b)}(\mathbf{x})$ denotes the terminal node of tree b to which observation $\mathbf{x}$ is routed, $\hat{H}^{(b)}(\cdot)$ is the Nelson–Aalen cumulative hazard estimate for that node, and B is the total number of trees. Averaging over B trees reduces the variance that any single tree would carry, and the corresponding survival function estimate is recovered via $\hat{S}(t | \mathbf{x}) = \exp(-\hat{H}(t | \mathbf{x}))$ as in Eq. (2.3).

RSF is different relative to the other models: like XGBoost-AFT, it captures non-linear interactions; like the Cox model, it operates within the survival framework and naturally handles censored data; and like the KM estimator, it is non-parametric in the sense that it makes no distributional assumptions about the shape of dwell times.

Feature importance is assessed via *Variable Importance* (VIMP), a permutation-based measure introduced alongside RSF by [15]. Because each tree is grown on a bootstrap sample, roughly one third of the training observations are never used for fitting that tree; these *out-of-bag* (OOB) observations form a natural held-out set for each tree. VIMP for a feature x_m is computed as follows: first, the OOB C-index is recorded using the original, unpermuted feature values; then, the values of x_m are randomly permuted across the OOB observations, breaking any true association between x_m and the outcome, and the C-index is re-evaluated. The VIMP of feature x_m is the resulting drop in OOB C-index:

$$\text{VIMP}_m = \hat{C}_{\text{OOB}} - \hat{C}_{\text{OOB}}^{\pi_m}, \quad (2.17)$$

where $\hat{C}_{\text{OOB}}$ is the out-of-bag concordance index before permutation and $\hat{C}_{\text{OOB}}^{\pi_m}$ is the concordance index after randomly permuting feature x_m . A large positive VIMP indicates that x_m strongly contributes to predictive accuracy; a VIMP near zero (or negative, due to sampling noise) indicates the feature carries little predictive signal. This measure is averaged over all B trees to reduce variance.

2.5 Evaluation Metrics for Survival Models

Evaluating survival models is more nuanced than evaluating standard regression or classification models. It can be measured using C-index and Brier score which are defined below:

2.5.1 Concordance Index

The concordance index (C-index) [16] is the survival analysis analogue of the area under curve (AUC). It measures the fraction of all comparable parcel pairs for which the model correctly ranks their dwell times:

$$C = \frac{\sum_{i \neq j} \mathbf{1}(\hat{\eta}_i > \hat{\eta}_j) \cdot \mathbf{1}(t_i < t_j) \cdot \delta_i}{\sum_{i \neq j} \mathbf{1}(t_i < t_j) \cdot \delta_i}, \quad (2.18)$$

where $\hat{\eta}_i$ is the model's risk score for parcel i and the sums are taken over comparable pairs (those where the shorter event time corresponds to an observed pickup). A C-index of 0.5 is chance; a value of 1.0 is perfect discrimination. In operational terms, a high C-index means the model can correctly identify which parcels will occupy a compartment longest, which is the core input required for a dynamic capacity reservation system: if we know parcel A is likely to be collected before parcel B , we can safely accept a new delivery into parcel A 's compartment before B is collected.

2.5.2 Brier Score and Integrated Brier Score

The Brier score at time t [17] measures the mean squared error between the predicted survival probability $\hat{S}(t | \mathbf{x}_i)$ and the binary indicator of whether parcel i was still in the locker at t , corrected for censoring through inverse probability of censoring weighting (IPCW):

$$\text{BS}(t) = \frac{1}{n} \sum_{i=1}^n \left[\frac{\hat{S}(t | \mathbf{x}_i)^2 \cdot \mathbf{1}(t_i \leq t, \delta_i = 1)}{\hat{G}(t_i)} + \frac{(1 - \hat{S}(t | \mathbf{x}_i))^2 \cdot \mathbf{1}(t_i > t)}{\hat{G}(t)} \right], \quad (2.19)$$

where $\hat{G}(t)$ is the Kaplan-Meier estimate of the censoring survival distribution. A score of 0 indicates perfect calibration; a naive model that always predicts 0.5 achieves 0.25. Because

single time-point evaluation can be sensitive to the choice of t , we integrate over the full observed time range to obtain the *integrated Brier score* (IBS):

$$\text{IBS} = \frac{1}{t_{\max} - t_{\min}} \int_{t_{\min}}^{t_{\max}} \text{BS}(t) dt, \quad (2.20)$$

approximated numerically over the grid of observed event times. The IBS is the primary calibration metric used for model comparison in Chapter 4, as it summarises how accurately the model’s predicted survival curves track observed pickup behaviour across the entire dwell time horizon.

2.5.3 Interpreting C-index and IBS Together

The C-index measures discrimination; whether the model correctly ranks pairs of parcels by expected dwell time. The IBS measures calibration; whether the predicted survival probabilities are accurate in absolute terms. A model can rank correctly but be poorly calibrated, or be well calibrated but unable to rank individual parcels. In a low-signal dataset where observable features explain only limited variance in individual pickup times, a C-index near 0.5 does not imply model failure: it implies that individual ranking is not achievable with the available data. The IBS still captures whether the aggregate survival curves are well-calibrated, which remains informative for understanding population-level pickup patterns (RQ1).

3 Method

This chapter describes how the work was carried out in sufficient detail for it to be replicated. The work was organised into four parts: data preparation, exploratory analysis and feature engineering, model implementation and training, and evaluation. Throughout the chapter, model definitions and mathematical notation introduced in Chapter 2 are referenced without re-derivation.

3.1 Data Preparation

3.1.1 Data Source

The dataset was provided by PostNord and contains approximately 100,000 parcel-level records drawn from around 100 service points across Sweden, covering a three-month observation window. Each row corresponds to a single parcel delivery event, recording the full lifecycle from booking the locker through delivery by the person to customer pickup. Table 3.1 summarises the features present in the raw dataset.

Table 3.1: Raw features in the company dataset

Feature	Type	Description
service_point_id	Categorical	Unique service point identifier
piece_id	Categorical	Unique parcel identifier
compartment_id	Categorical	Compartment assigned to the parcel
booking_date	Datetime	Timestamp the locker was booked
parcel_delivery_date	Datetime	Timestamp of delivery to locker
parcel_pickup_date	Datetime	Timestamp of customer pickup
compartment_size	Categorical	Size class of compartment (S/M/L/S or M Slim)
piece_size	Categorical	Size class of parcel (S/M/L/S or MSlim)

3.1.2 Defining the Survival Outcome

The survival outcome is *dwelt time*: the elapsed time in hours between parcel delivery and customer pickup. This is computed as:

$$T_i = \text{parcel_pickup_date}_i - \text{parcel_delivery_date}_i \quad (\text{in hours}). \quad (3.1)$$

The event indicator $\delta_i = 1$ for all parcels with an observed pickup timestamp, and $\delta_i = 0$ for parcels that had not been collected at the time of data extraction (right-censored observations). A small fraction of parcels; less than 1% of the dataset; remain uncollected beyond seven days and are returned to sender by company policy. These are treated as right-censored at 168 hours ($T_i = 168$, $\delta_i = 0$) rather than as failed pickups, since their eventual pickup time is simply unknown.

The processed columns T (dwelt time in hours) and E (event indicator, 1 = pickup observed) are the inputs to all survival models.

3.1.3 Train-Test Split

The dataset is split into a training set (80%) and a held-out test set (20%). The split is performed by stratifying on `service_point_id`, so that each service point contributes proportionally to both partitions. This prevents data leakage from service-point level patterns: a model should not be evaluated on the same locker locations it was trained on without having seen any new locations. Model selection and hyperparameter tuning are performed entirely on the training set using 5-fold cross-validation (see Section 3.4), and the test set is reserved for final evaluation in Chapter 4.

3.2 Exploratory Analysis and Feature Engineering

3.2.1 Dwell Time Distribution

Preliminary analysis of the dwell time variable shows a heavily right-skewed distribution. The median dwell time is approximately 20 hours, with 56.3% of parcels collected within 24 hours. However, the distribution has a pronounced long tail: around 8% of parcels remain in their compartment beyond 72 hours, and some remain for the full seven-day retention period. The empirical survival curve confirms this structure, with $\hat{S}(24\text{h}) \approx 0.43$ and $\hat{S}(72\text{h}) \approx 0.08$. These summary statistics signify the heavy right tail; which is the primary operational problem. Parcels in the tail occupy compartments for long periods, reducing effective capacity and driving assignment failures. This can be seen the figure demonstrating the empirical survival curve, which has a steep initial decline followed by a long, flat tail.

The right-skewed shape directly motivates the parametric model choice in Section 3.3: the Weibull and Log-Normal distributions are both well-suited to right-skewed, positive-valued data with decreasing hazard, while symmetric or unbounded distributions such as the normal would be inappropriate.

3.2.2 Feature Engineering

The following features were derived from the raw timestamps and included in the model features:

- **delivery_hour**: the hour of day (0-23) at which the parcel was delivered. Deliveries in the early morning are expected to have different pickup patterns from those arriving in the afternoon. This feature was included to see within-day pickup patterns.
- **delivery_dayofweek**: integer day of the week (0 = Monday, 5 = Friday), capturing the weekend effect.

- **delivery_month**: calendar month, capturing seasonal variation.
- **booking_to_delivery_hours**: the elapsed time between the booking date and the delivery date. This feature was included just to validate any correlation between booking time and delivery time and its effect on the pickup.
- **size_match**: a binary indicator equal to 1 if the parcel size class matches the compartment size class exactly, and 0 if the parcel was placed in an oversized compartment. A mismatch could indicate sub-optimal assignment and could be an indicator of operational inefficiencies.

Two features that were initially created were subsequently dropped after careful inspection of the data. **is_weekend_delivery** was dropped because the dataset contains no deliveries on Saturdays or Sundays; all deliveries occur on weekdays, reducing this binary feature to identically zero with no predictive value. **delivery_hour_bucket** (a version of **delivery_hour**) was dropped because it provided no additional information beyond what the raw hour variable already captures. The initial thoughts behind this feature was to see if there are patterns in specific hours of the day in the form of buckets. However, it did not provide better signal than the hour variable.

Pickup timestamps were never used as input features to the models, to prevent data leakage. Categorical features were one-hot encoded for the Cox PH and Parametric AFT models, and ordinally encoded for tree based models i.e. Random Survival Forests and XGBoost-AFT.

The final feature matrix used for covariate models contained the following columns: **delivery_hour**, **delivery_dayofweek**, **delivery_month**, **compartment_size**, **piece_size**, **booking_to_delivery_hours**, and **size_match**.

3.2.3 Stratified Kaplan-Meier Analysis

Before fitting any covariate model, stratified Kaplan-Meier curves were computed to provide a visual, assumption-free inspection of how the main categorical covariates relate to dwell time. Curves are computed separately for:

- **Piece size** (**piece_size**): to assess whether parcel size class affects pickup speed independently of compartment size.
- **Delivery day of week** (**delivery_dayofweek**): to assess the Friday-to-Monday effect, where parcels delivered late in the working week are expected to have longer dwell times due to weekend effect.
- **Delivery month** (**delivery_month**): to detect seasonal variation within the observation window.

The KM curves serve as a sanity check on the dataset's behavioural signal. A dataset where stratified curves are indistinguishable across all covariates would indicate that the covariates carry no information about pickup timing, and none of the covariate models would be expected to outperform the marginal KM baseline.

In the figures 4.5, 4.3 you can see all of these stratified curves. The curves show clear stratification by delivery day of week and piece size, with parcels delivered on Fridays having the longest dwell times and those delivered on Mondays having the shortest. The piece size curves show some stratification, with larger parcels tending to have longer dwell times, but the differences are less pronounced than for the day of the week curve.

3.3 Model Implementation

Four types of models are fitted following the survival analysis framework described in Chapter 2. The models are intended to address the research questions at increasing levels of complexity: KM for population-level baseline (RQ1), Cox PH and Parametric AFT (Weibull and Log-Normal) for covariate-adjusted distributional modelling (RQ1, RQ2), and Machine Learning (RSF) for non-linear feature importance analysis (RQ2).

3.3.1 Kaplan-Meier Estimator

The Kaplan-Meier estimator (Section 2.2.1) is fitted on the full training set to establish the marginal pickup time distribution. It is additionally fitted on subgroups defined by `piece_size` and `delivery_dayofweek` to confirm that the stratification signals identified in Section 3.2.3 persist under the full dataset. Log-rank tests are used to assess whether the stratified curves differ statistically. These tests were conducted for clarification purposes. The survival curves referred in Section 3.2.3 already signify that there is significant difference in the segments. The KM estimator does not generate predictions for the test set and is not included in the comparative evaluation of Section 3.4; its role is purely descriptive and used for sanity checks.

3.3.2 Cox Proportional Hazards Model

The Cox PH model (Section 2.2.2) was fitted using the `lifelines` library (`CoxPHFitter`) with an ℓ_2 regularisation penalty (`penalizer = 0.1`). This value was selected based on the grid search results on `penalizer` and `l1_ratio` parameters; which is discussed further briefly in this section. All covariates entered the model linearly on the log-hazard scale.

After fitting, the proportional hazards assumption is verified for each covariate using Schoenfeld residuals [8]. Any covariate that shows a statistically significant time-varying effect ($p < 0.05$) is either stratified on or modelled with a time-interaction term. This verification step is essential because the Cox model produces invalid inference if the PH assumption is materially violated.

Per-parcel survival curves on the test set are obtained by combining the estimated $\hat{\beta}$ with the Breslow estimator of the baseline cumulative hazard, as described in Section 2.2.2.

Grid search was done on the hyperparameters available for the Cox PH model. The parameters used alongwith their values can be seen in the table 3.2 below. The `penalizer` combines both ℓ_1 and ℓ_2 penalties, with the relative weight determined by `l1_ratio` parameter in the function. Finally, stratification effects were also taken into account on the `'delivery_dayofweek'` and `'piece_size_enc'` variables.

Table 3.2: Hyperparameters for Cox PH

Hyperparameter	Values
<code>penalizers</code>	[0.0, 0.01, 0.1, 1.0, 10.0]
<code>l1_ratios</code>	[0.0, 0.5, 1.0]
<code>method</code>	[<code>'breslow'</code>]
<code>strata_options</code>	[<code>None</code> , [<code>'delivery_dayofweek'</code>], [<code>'piece_size_enc'</code>],
-	[<code>'delivery_dayofweek'</code> , <code>'piece_size_enc'</code>]]

An attempt was also made to fit the Cox model with splines but due to the data size and the computational cost, it was not possible to produce C-index scores. Although the model produced similar results to the Breslow estimator. The results did not show any significant differences hence were skipped in the final evaluation.

3.3.3 Parametric AFT Models

Two parametric AFT families are fitted and compared: Weibull and Log-Normal. Both are implemented using `lifelines` (`WeibullAFTFitter` and `LogNormalAFTFitter`). The log-linear covariate structure of Eq. (2.9) is used unchanged - no spline terms were added, as the log-linear AFT parameterisation places the covariate effect directly on $\log T$ rather than on the hazard, and the relationship tends to be well-behaved for the features in this dataset.

Both models are fitted by maximising the full log-likelihood of Eq. (2.4) with their respective parametric forms for $f(t)$ and $S(t)$. Model selection between the two families is performed on the training set using the Akaike Information Criterion (AIC), the equation for the same can be found here Eq. (2.13)

The lower-AIC model is preferred for the final parametric model. Both are reported in the Chapter 4, as the AIC difference carries information about the distribution shape that fits the data.

The acceleration factors $\exp(\hat{\beta}_j)$ for each covariate j are reported as the primary interpretable output for RQ2, alongside their 95% confidence intervals obtained from the observed Fisher information matrix.

Grid Search was carried out on the hyperparameters of both the models. Similar to 3.2, the parameters used alongside their values can be seen in the table 3.3 below. The penalizer combines both L_1 and L_2 penalties, with the relative weight determined by `l1_ratio`.

Table 3.3: Hyperparameters for AFT models

Hyperparameter	Values
<code>penalizers</code>	[0.0, 0.01, 0.1, 1.0, 10.0]
<code>l1_ratios</code>	[0.0, 0.5, 1.0]

3.3.4 Random Survival Forest

The RSF (Section 2.4.2) was implemented using the `scikit-survival` library (`RandomSurvivalForest`). The following fixed hyperparameters were used:

- **Number of trees:** 100. This is sufficient for stable out-of-bag estimates and permutation importance scores without incurring excessive computational cost.
- **Features per split (`max_features`):** $\sqrt{p}$, where p is the number of features, following the standard recommendation for survival forests in [15].
- **Minimum node size (`min_samples_leaf`):** 15. This prevents extremely small terminal nodes, which would produce noisy cumulative hazard estimates and avoid overfitting.
- **Parallelism (`n_jobs`):** set to 1. The dataset contains over 80,000 unique event times, which causes memory exhaustion when multiple trees attempt simultaneous cumulative hazard estimation. Running trees sequentially avoids this out of memory problem.
- **Dwell time rounding:** event times are rounded to the nearest integer hour before fitting. This reduces the number of unique event times from ~82,000 to ~336 (14 days $\times$ 24 hours), that reduces memory consumption while preserving the resolution relevant for operational decisions.

Due to the time constraints and the amount of time the model takes to train, no hyperparameter tuning was done for the RSF model. The selected hyperparameters are based on standard recommendations from the best practices and preliminary experiments on a smaller subset of the data.

Feature importance was assessed in two complementary ways. *Permutation-based variable importance* (VIMP) measures the increase in out-of-bag prediction error when the values of a given feature are randomly permuted across observations, and the C-index was recorded as importance score. A positive VIMP score indicates that the feature carries predictive information [15]. *SHAP values* [18] are computed using a tree-based SHAP estimator on the fitted forest, providing instance-level attribution that reveals not just which features matter globally, but in which direction and for which parcels. This dual importance analysis forms the primary answer to RQ2.

3.3.5 XGBoost-AFT Model

The XGBoost-AFT model was implemented using the native `survival:aft` objective in `xgboost` [14] with the log-normal error distribution (`aft_loss_distribution = 'normal'`), consistent with the Log-Normal AFT parametric baseline and enabling direct comparison.

Survival curves were reconstructed from the model’s predicted log-time output $\hat{\mu}_i$ and an empirical scale $\hat{\sigma}$ estimated from training residuals:

$$\hat{S}(t \mid \mathbf{x}_i) = 1 - \Phi\left(\frac{\log t - \hat{\mu}_i}{\hat{\sigma}}\right), \quad (3.2)$$

where $\Phi(\cdot)$ is the standard normal CDF. This reconstruction step is necessary because XGBoost-AFT outputs $\hat{\mu}_i$ (the predicted value of $\log T$) directly rather than a hazard ratio, so no Breslow estimator is required.

3.4 Evaluation

3.4.1 Evaluation Metrics

Models were evaluated on the held-out test set using the metrics defined in Section 2.5: the concordance index (C-index), the Integrated Brier Score (IBS). Table 3.4 summarises what each metric measures and interpretation.

Table 3.4: Summary of evaluation metrics

Metric	Measures	Good value	Reference
C-index	Discrimination (Pairwise Ranking)	Closer to 1.0	[16]
IBS	Calibration (Survival Curves)	Closer to 0	[17]

3.4.2 Cross-Validation and Model Comparison

For the Cox PH, Weibull AFT, Log-Normal AFT, hyperparameter selection was performed via 5-fold cross-validation on the training set, stratified by `service_point_id`. The best configuration per model was retrained on the full training set before test-set evaluation. Results across all six models were reported in a single comparison table covering C-index, IBS, and where applicable the AIC from training.

3.4.3 Low Signal: Expectation and Interpretation

Grid search revealed little improvement over default hyperparameters across all models: all C-index values on the validation folds converged in the range 0.64-0.66, well above chance but comparable in absolute terms. This pattern is consistent with a *low-signal dataset*, where the available features explain only a limited portion of the variance in individual pickup times. The remaining variance is likely attributable to unobservable factors such as personal schedule, importance of the parcel, weather conditions, and distance from the service point.

In such a setting, a C-index of ~ 0.65 does not indicate models are not trained well; it is a reflection of the information content of the feature set. The IBS provides complementary evidence: a well-calibrated model can still produce accurate aggregate survival curves even when individual ranking is not precise, which remains operationally useful for understanding population-level pickup patterns (RQ1).

4 Results and Discussion

This chapter presents the results of the analyses described in Chapter 3 and discusses their meaning in relation to the two research questions. To avoid separating findings from their interpretation, results and discussion are presented together: each section reports the outcome of one phase of the analysis and then discusses what that outcome implies. The chapter follows the same four-phase structure as Chapter 3.

4.1 Exploratory Analysis

4.1.1 Dwell Time Distribution

Figure 4.1 shows the distribution of parcel dwell times. The distribution is strongly right-skewed with a median of 20.7 hours. Most parcels are collected within the first day, producing a mode in the 5-15 hour range. Beyond 24 hours, the distribution thins progressively through a long right tail. The spike at 168 hours is a censoring effect: parcels that exceed PostNord’s seven-day retention window are returned to sender and treated as right-censored at 168 hours in all survival models.

The overall Kaplan-Meier survival curve in Figure 4.2 quantifies this pattern. The curve crosses $S = 0.5$ at the median dwell time of 20.6 hours. By 72 hours, only 8% of parcels remain in the locker. Table 4.1 reports the key survival probabilities.

Table 4.1: Key dwell time summary statistics from the overall Kaplan-Meier estimate

Statistic	Value	Interpretation
Median dwell time	20.7h	Half of all parcels collected within ~21h
$\hat{S}(24h)$	0.43	57% collected within one day
$\hat{S}(72h)$	0.08	92% collected within three days
$\hat{S}(168h)$	0.00	All parcels resolved within seven days

The shape of the dwell time distribution directly addresses RQ1 at the population level. The steep initial decline confirms that the majority of customers behave promptly, and the dominant driver of locker occupancy is a minority of slow-pickup parcels: the 8% that remain beyond 72 hours occupy compartments for disproportionately long periods and are the primary source of capacity pressure. This long-tail structure is consistent with findings from

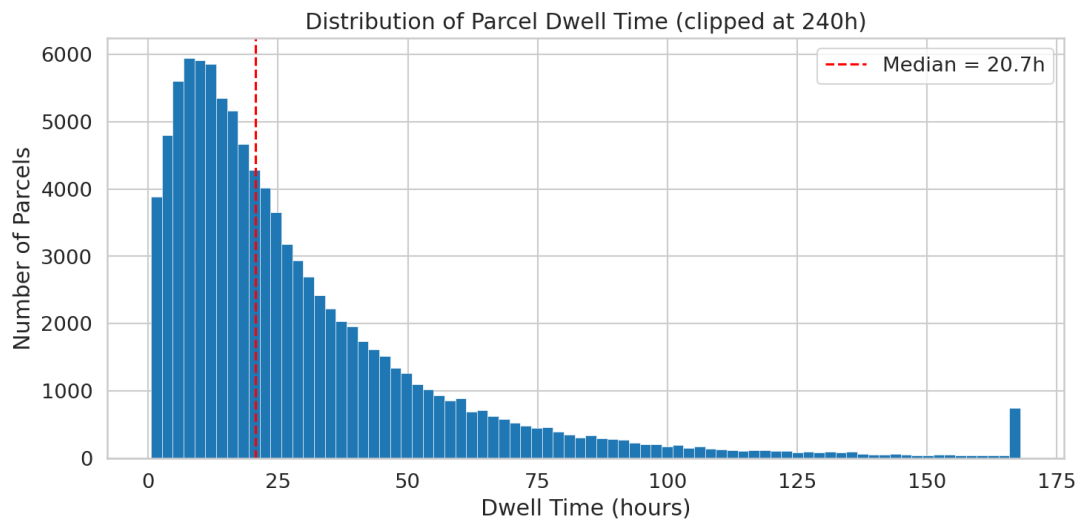


Figure 4.1: Distribution of parcel dwell times

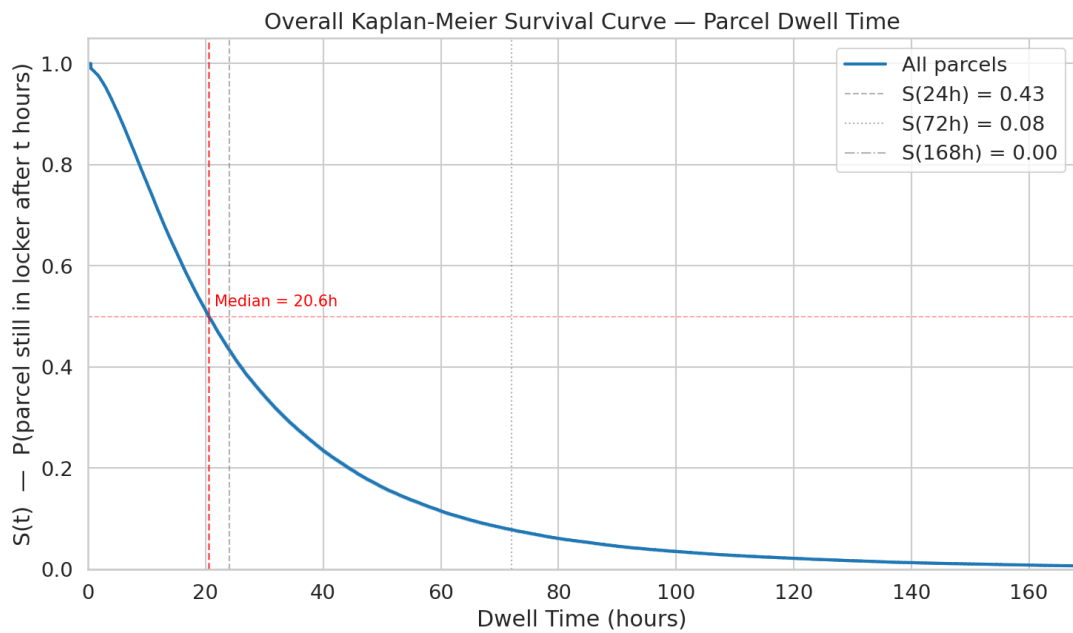


Figure 4.2: Overall Kaplan-Meier survival curve

deployed locker networks in the United States, where a similar minority of delayed pickups was found to substantially reduce daily throughput capacity [1]. The implication for operations is that targeted interventions aimed at the slow-pickup segment; such as escalating notification reminders after 48 hours; are likely to have a larger impact on capacity than interventions aimed at the median customer.

4.1.2 Stratified Kaplan–Meier Analysis

Delivery day of week: Figure 4.3 shows the most visible differences in the dataset. Monday deliveries have the shortest dwell times (median $\approx 16\text{h}$), while Friday deliveries are dramatically slower (median $\approx 39\text{h}$), as summarised in Figure 4.4. The Friday survival curve remains visibly separated from all weekday curves across the entire 168-hour window.

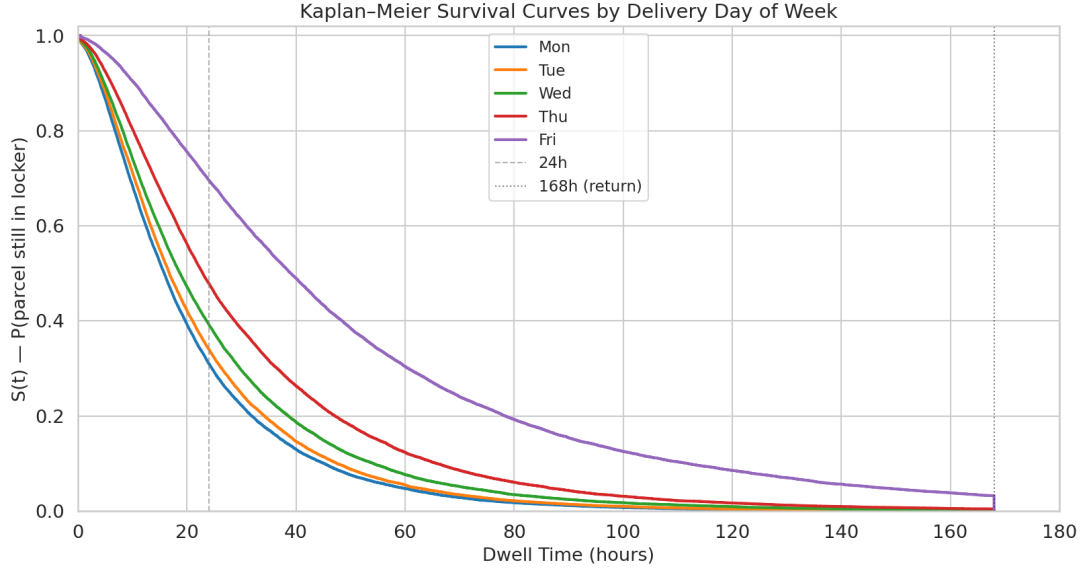


Figure 4.3: Kaplan-Meier survival curves by delivery day of week

Piece size: Figure 4.5 shows a clear monotone ordering by parcel size class, from small (median 14.9h, $S(24\text{h}) = 0.30$) to large (median 32.5h, $S(24\text{h}) = 0.63$). Table 4.2 summarises the group-level statistics.

Table 4.2: KM summary statistics by piece size group (training set)

Piece size	Median (h)	$\hat{S}(24\text{h})$
Small	14.9	0.295
Small slim	16.6	0.337
Medium	20.9	0.436
Medium slim	23.0	0.482
Large	32.5	0.627

Delivery hour: Figure 4.6 shows that afternoon deliveries (hours 15-17) are associated with median dwell times roughly 9-10 hours longer than morning deliveries (hours 8-11), spanning 17-18h versus 26-27h respectively. Afternoon deliveries arrive when customers are unlikely to visit the service point again until the following day, explaining the elevated dwell times.

The stratified KM results provide a model-free, assumption-free answer to RQ2 at the group level. Three features carry visually unambiguous signal before any regression model

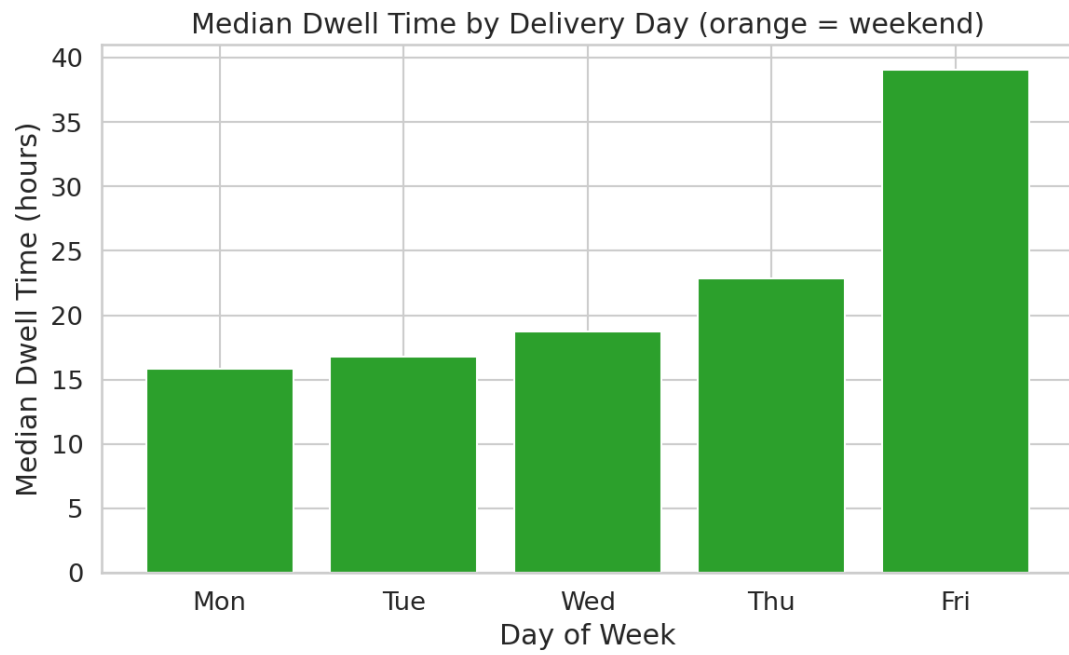


Figure 4.4: Median dwell time by delivery day of week

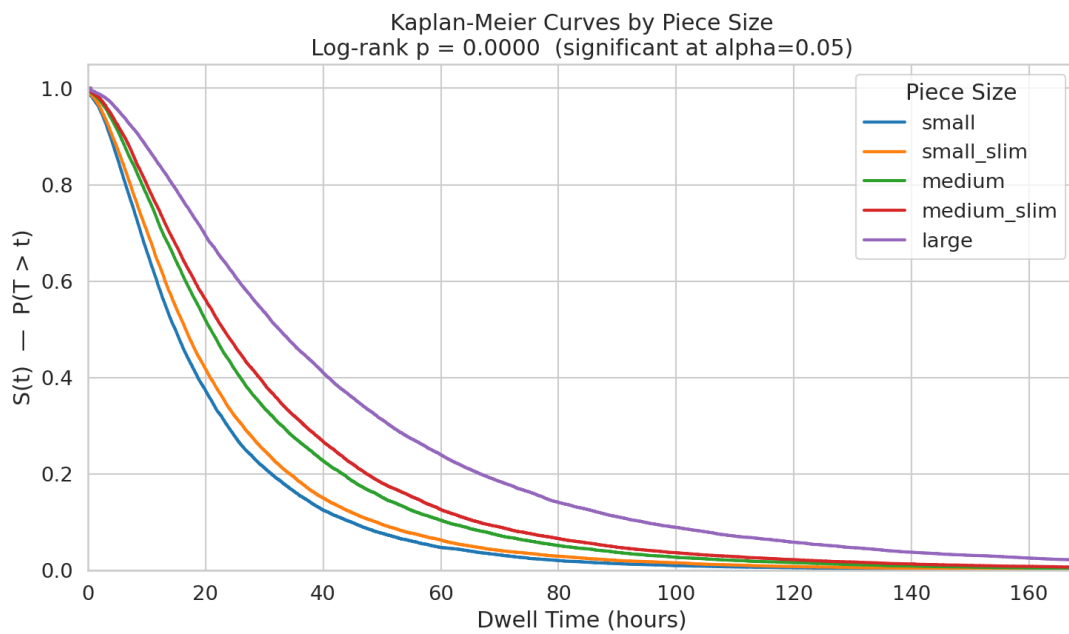


Figure 4.5: Kaplan-Meier survival curves by piece size

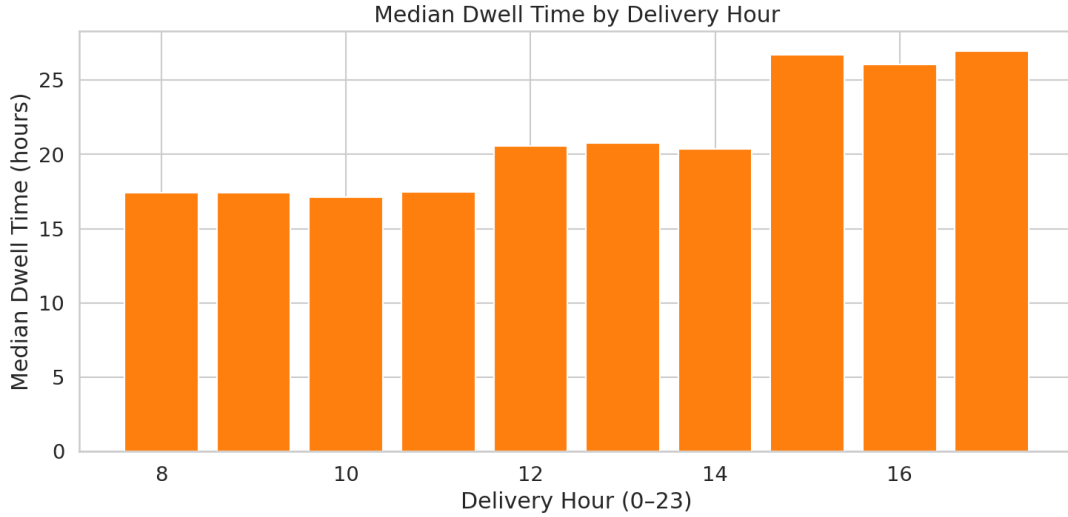


Figure 4.6: Median dwell time by delivery hour

is fitted: delivery day of week, piece size, and delivery hour. The day-of-week effect is the strongest, and it is transparent: no deliveries occur on weekends, so a Friday parcel waits at minimum 48-60 hours before the customer's first post-weekend opportunity to collect. The piece size ordering is also consistent with an intuitive explanation; larger parcels require more effort to carry, reducing the urgency or convenience of collection.

The log-rank p -values are all reported as $p < 0.0001$, which is expected given the large sample size ($n \approx 84,000$ training observations). Statistical significance at this scale is almost guaranteed for any non-trivial effect; the practical importance of each factor is better judged from the KM curves and median dwell times than from the p -values alone.

4.2 Model Performance

As seen in the 4.7 the C-index values are constant throughout the dwell time period from 0 to 168 hours, the XGBoost-AFT model again outperforms the other models.

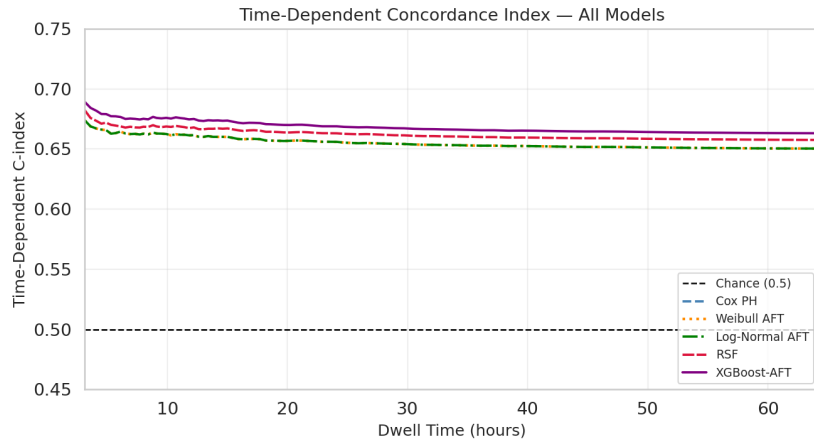


Figure 4.7: Time Dependent C-index for all models: 10th to 90th Percentiles

The 4.8 shows the Brier Score for all models over the time axis. As observed the Brier Score is increasing over time till ~ 20 hours, then decreasing after that. This is because of the uncertainty of the model prediction during the median hour. The models are uncertain as most of the pickups happen around this time. The decrease after this is also explainable as most of the parcels are picked up and the models predict that the survival of parcels is approaching zero, which is easier to predict. In this case also, we can see that the models are very much close and XGBoost-AFT is performing better than the other models by a slight margin.

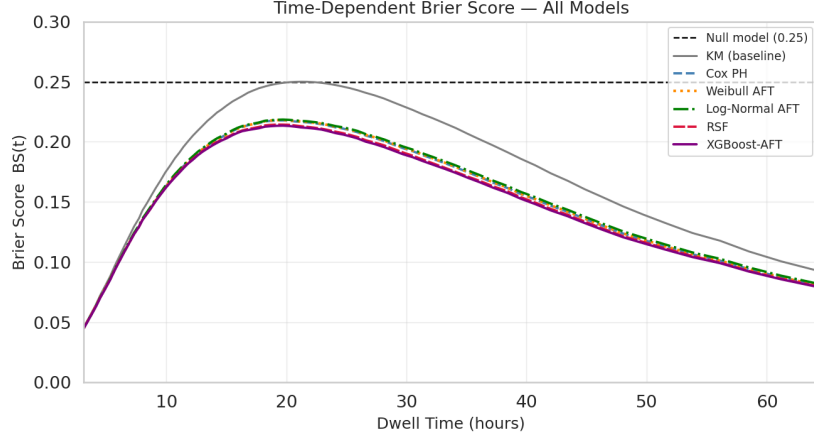


Figure 4.8: Time Dependent Brier Score for all models: 10th to 90th Percentiles

Table 4.3: Model performance on the held-out test set

Model	C-index	IBS	AIC (train)
Cox PH	0.6499	0.1529	—
Weibull AFT	0.6500	0.1531	713,244
Log-Normal AFT	0.6500	0.1542	721,995
RSF	0.6597	0.1507	—
XGBoost-AFT	0.6627	0.1498	—

Table 4.3 reports the C-index and IBS for all models on the held-out test set ($n = 16,041$ parcels). Two observations stand out from Table 4.3.

First, all models with covariates achieve C-index values in the narrow range 0.649-0.663, with a total spread of only 0.013 points across five models of very different complexity. This convergence is the defining empirical finding of this thesis: the available features; delivery day, delivery hour, parcel size, compartment size, and size match; carry limited discriminative information about *individual* pickup ordering. Increasing model complexity from a linear Cox model to a complex XGBoost ensemble yields a gain of only 0.013 C-index points, which is negligible in operational terms. The primary explanation is that the most important drivers of when a specific customer collects their parcel; personal schedule, distance from the service point, awareness of the notification, and parcel priority; are not recorded in the data. This reflects the information content of the available feature set, consistent with the low-signal diagnosis described in Section 3.4.3.

Second, both RSF (IBS = 0.1507) and XGBoost-AFT (IBS = 0.1498) produce survival curves that are substantially better calibrated than the null reference of 0.25, and XGBoost-AFT achieves the best calibration overall. This is a meaningful result for RQ1: even when individual pickup ordering cannot be predicted precisely, the survival models correctly capture the shape and timescale of the pickup time distribution at the population level.

The Weibull AFT model is strongly preferred over Log-Normal on distributional fit grounds ($\Delta\text{AIC} = 8,751$), despite having virtually identical discrimination performance.

4.3 Cox Proportional Hazards Model

4.3.1 Hyperparameter Selection

Grid search over penalizer $\in \{0, 0.01, 0.1, 1.0, 10.0\}$ and ℓ_1 -ratio $\in \{0.0, 0.5, 1.0\}$ via 5-fold cross-validation selected **penalizer** = 0.01, **l1_ratio** = 1.0. Figure 4.9 shows the heatmap of mean cross-validation C-index values across the grid. The selection of configuration was although done by lower standard deviation on l_1 ratio as it is more stable across folds.

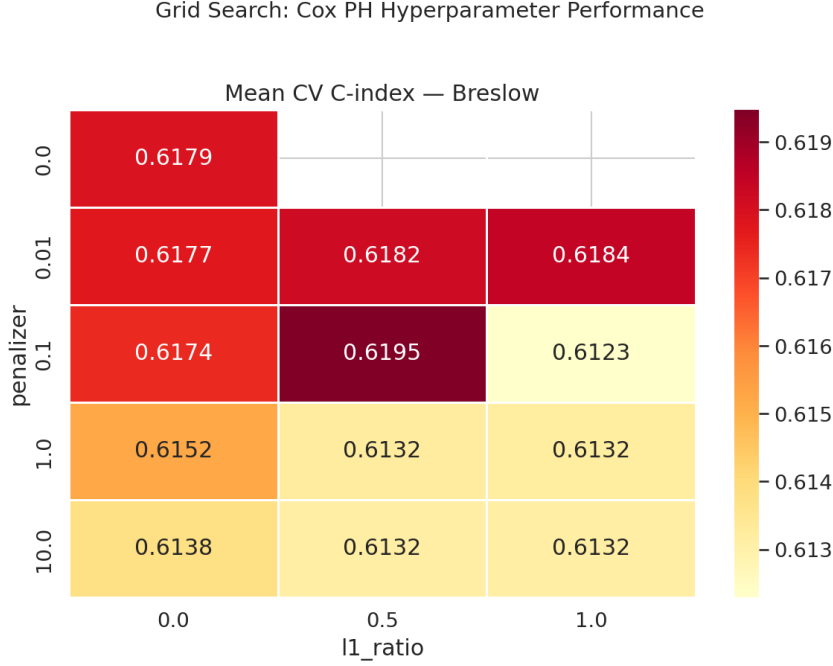


Figure 4.9: Mean cross-validation C-index for the Cox PH model

4.3.2 Coefficient Estimates and Proportional Hazards Check

Table 4.4 reports the fitted hazard ratios. All covariates except **booking_to_delivery_hours** are statistically significant ($p < 0.05$). There was one interesting thing to observe in this table, the CI for every variable is very tight, this was because of the large sample size.

$\exp(\hat{\beta}) > 1$ indicates an increased instantaneous pickup rate (faster collection) while $\exp(\hat{\beta}) < 1$ indicates a decreased rate.

The Schoenfeld residual test [8] flagged five of seven covariates as violating the proportional hazards assumption, as shown in Table 4.5.

The PH violations are concentrated in precisely the features that carry the strongest predictive signal. This pattern was expected and it reflected a genuine operational phenomenon. The effect of delivery day of week on pickup hazard is not constant over time as a parcel delivered on a Friday has a near-zero hazard during the weekend hours but a high hazard on Monday morning, a pattern that cannot be captured by a single time-invariant hazard ratio. Similarly, delivery hour and parcel size may influence pickup behaviour differently in the first 24 hours (when most pickups occur) versus the long tail beyond 72 hours.

Table 4.4: Cox PH coefficient estimates

Covariate	$\hat{\beta}$	$\exp(\hat{\beta})$	95% CI	p
delivery_hour	-0.071	0.931	[0.926, 0.931]	<0.001
delivery_dayofweek	-0.252	0.777	[0.773, 0.781]	<0.001
delivery_month	-0.010	0.990	[0.980, 0.999]	0.028
booking_to_delivery_hrs	<0.001	1.000	[0.999, 1.000]	0.946
size_match	-0.091	0.913	[0.893, 0.929]	<0.001
compartment_size_enc	+0.168	1.183	[1.162, 1.191]	<0.001
piece_size_enc	+0.112	1.119	[1.106, 1.126]	<0.001

Table 4.5: Schoenfeld residual test for the proportional hazards (PH) assumption

Covariate	Statistic	PH
booking_to_delivery_hours	0.643	Yes
delivery_month	1.772	Yes
compartment_size_enc	14.981	No
delivery_dayofweek	8.414	No
delivery_hour	44.123	No
piece_size_enc	33.599	No
size_match	5.039	No

The practical consequence is that the Cox model’s hazard ratios in Table 4.4 should be interpreted as average effects across the entire dwell time horizon rather than as precise estimates at any specific time point. The AFT models of Section 4.4 and the machine learning models of Section 4.5 do not require the PH assumption and were therefore considered more appropriate for this dataset and further implemented.

4.4 Parametric AFT Models

4.4.1 Model Selection and Shape Parameter

Table 4.6 compares the two parametric models and to choose the better one.

Table 4.6: Weibull vs Log-Normal AFT model comparison

Model	C-index (test)	AIC (train)
Weibull AFT	0.6500	713,244
Log-Normal AFT	0.6500	721,995
Δ AIC		8,751

The AIC difference of 8,751 in favour of Weibull provides evidence that the Weibull distributional form better captures the shape of dwell times in this dataset. The estimated Weibull shape parameter is $\hat{\rho} = 1.3763$ which is defined as k in 2.3.2 and equation 2.11.

4.4.2 Acceleration Factors

Table 4.7 reports the Weibull AFT acceleration factors. An acceleration factor above 1 means a one-unit increase in the covariate is associated with longer expected dwell time; below 1 means shorter. For example: A value of 1.22 indicates a 22% longer expected dwell time per unit increase in the covariate.

The Weibull shape parameter $\hat{\rho} = 1.3763$ deserves careful attention. A value above 1 implies an *increasing* hazard: the instantaneous pickup rate grows over time. This is the opposite of the decreasing hazard hypothesised in Section 2.3.2 based on the expectation that customers who have not collected promptly become progressively less likely to do so. The increasing hazard

Table 4.7: Weibull AFT acceleration factors ($\exp(\hat{\beta})$)

Covariate	$\hat{\beta}$	$\exp(\hat{\beta})$	p
delivery_dayofweek	+0.202	1.224	<0.001
size_match	+0.074	1.077	<0.001
delivery_hour	+0.057	1.059	<0.001
delivery_month	+0.009	1.009	0.018
piece_size_enc	-0.090	0.914	<0.001
compartment_size_enc	-0.134	0.875	<0.001
booking_to_delivery_hours	<0.001	1.000	0.943

may reflect a reminder-driven pickup pattern: customers who do not collect immediately may do so in response to a notification or reminder sent by the company after a delay, which would produce an elevated pickup rate at moderate dwell times. Alternatively, it may be also an effect caused by the truncation at 168 hours: all parcels are picked up by day seven (right censored), which compresses the right tail and can inflate the estimated shape parameter.

The acceleration factors tell a coherent operational story. Each additional day of the week (Monday through Friday) stretches the expected dwell time by a factor of 1.22, which is the single largest time-ratio effect. This translates directly to the Friday effect observed in the KM analysis. The `booking_to_delivery_hours` feature is consistently non-significant across all models ($p = 0.943$), confirming that lead time carries no independent predictive information about when a customer will collect their parcel.

4.5 Machine Learning Models

4.5.1 Random Survival Forest

The RSF achieved a test C-index of 0.6597 and an IBS of 0.1527. Table 4.8 reports permutation VIMP scores for all features. A higher VIMP indicates a larger drop in predictive accuracy when the feature is permuted.

Table 4.8: RSF permutation variable importance (VIMP). Baseline C-index = 0.6597

Covariate	VIMP	Rank
delivery_dayofweek	0.0777	1
compartment_size_enc	0.0272	2
delivery_hour	0.0201	3
piece_size_enc	0.0154	4
size_match	0.0007	5
booking_to_delivery_hours	0.0001	6
delivery_month	0.0000	7

The RSF VIMP scores confirm and extend the KM findings. `delivery_dayofweek` dominates by a wide margin: its VIMP of 0.0777 is approximately three times larger than the next-ranked feature. This means that if we remove knowledge of which day of the week a parcel was delivered, the model's ability to rank parcels by expected dwell time drops by 7.8 percentage points; a substantial fraction of the total discrimination above chance. `delivery_month` and `booking_to_delivery_hours` both show VIMP scores at or near zero, providing convergent evidence from a non-parametric model that these features contribute nothing to predictive accuracy.

The training C-index of RSF (0.684) is higher than the test C-index (0.6597), a gap of 0.024 points. Given that no hyperparameter tuning was performed for RSF, this degree of overfitting is expected: the 100-tree forest can memorise local patterns in the training data that do not generalise. Tuning `min_samples_leaf` and `max_features` on the validation folds

would likely reduce this gap, though the small magnitude of the gap suggests the model has not overfit severely.

4.5.2 XGBoost-AFT

4.5.2.1 Hyperparameter Selection

A grid search over 729 hyperparameter combinations (5-fold cross-validation, maximum 1,500 boosting rounds with early stopping at 30 rounds) identified the configuration in Table 4.9.

Table 4.9: XGBoost-AFT best hyperparameters from 5-fold cross-validation over 729 combinations

Hyperparameter	Selected value
max_depth	3
learning_rate	0.01
subsample	0.8
min_child_weight	10
reg_alpha (ℓ_1)	1.0
reg_lambda (ℓ_2)	2.0
Boosting rounds	761

The selection of a shallow tree depth (`max_depth` = 3) combined with a low learning rate and strong regularisation ($\ell_1 = 1.0$, $\ell_2 = 2.0$) is consistent with the low-signal nature of the dataset: the optimal configuration is one that avoids overfitting to noise rather than one that builds complex interaction trees.

4.5.2.2 Predictive Performance

The XGBoost-AFT model achieved the best C-index of all models (0.6627) and the best IBS (0.1498), with an estimated scale parameter $\hat{\sigma} = 0.9154$. The training loss decreased from 10.78 at round 0 to 4.26 at round 658, with diminishing returns beyond approximately 150 rounds in the validation set achieving the lowest loss of 4.37. Early stopping at 30 rounds after the minimum was triggered at round 658 and best iteration was found at round 628.

4.5.2.3 SHAP Feature Importance

Figures 4.10 and 4.11 show the SHAP feature importance bar chart and beeswarm plot respectively. Table 4.10 reports the mean absolute SHAP values. `delivery_dayofweek` has roughly 1.5 times the influence of `compartment_size_enc`, the second-ranked feature. The bottom three features contribute negligible impact.

Each point in 4.11 is a test-set observation. Colour indicates feature value (red = high, blue = low); horizontal position indicates the SHAP contribution to predicted log-dwell-time. Positive SHAP values push the prediction toward longer dwell times; negative values push it shorter. Higher values indicate greater average impact on the predicted log-dwell-time $\hat{\mu}_i$.

Table 4.10: XGBoost-AFT mean absolute SHAP values on the test set

Covariate	Mean SHAP	Rank
delivery_dayofweek	0.261	1
compartment_size_enc	0.170	2
delivery_hour	0.148	3
piece_size_enc	0.137	4
size_match	0.013	5
booking_to_delivery_hours	0.004	6
delivery_month	0.003	7

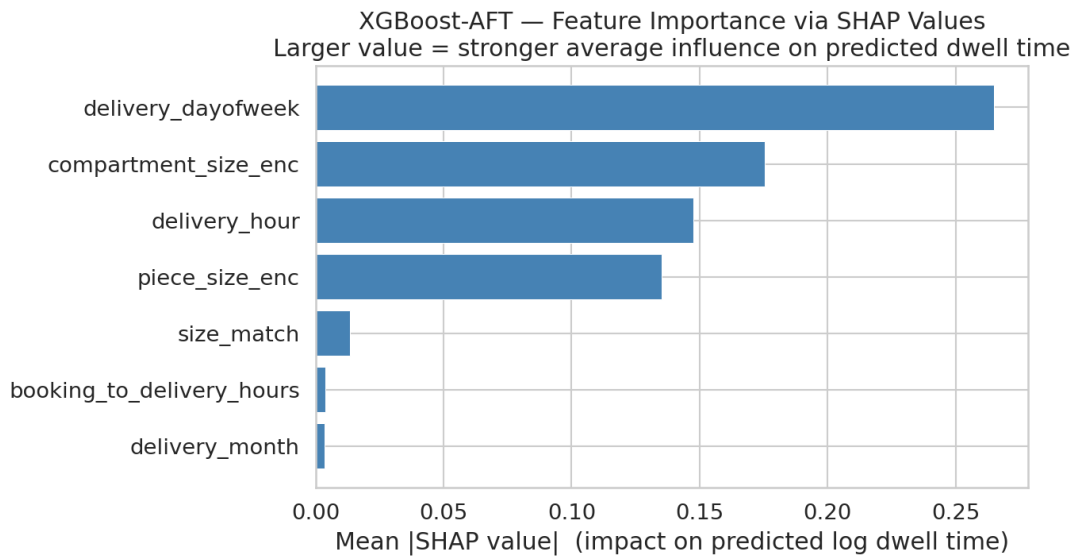


Figure 4.10: XGBoost-AFT mean absolute SHAP values

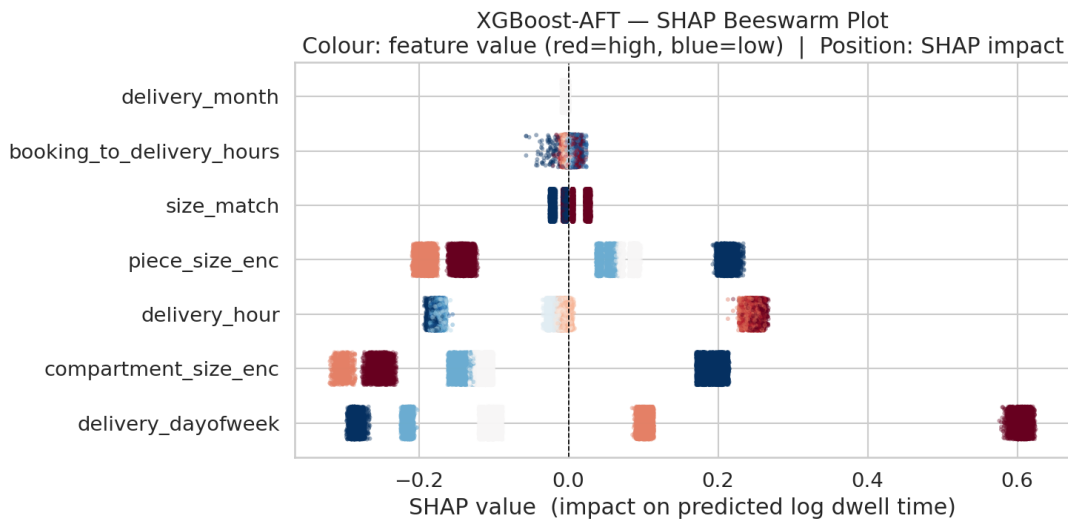


Figure 4.11: SHAP beeswarm plot

The beeswarm plot in Figure 4.11 reveals the direction and non-linearity of each feature's effect in a way that the bar chart and coefficient tables cannot. For `delivery_dayofweek`, there is a pronounced cluster of high-value (red) observations with SHAP values exceeding +0.6, corresponding to Friday deliveries that push predicted dwell time sharply upward. This confirms that the day-of-week feature acts primarily through the Friday effect rather than through a gradual linear trend across all weekdays.

The `delivery_hour` beeswarm shows a bimodal pattern: early deliveries (blue, low hour values) cluster at slightly negative SHAP values, while late-afternoon deliveries (red, high hour values) spread toward positive SHAP values. This directional pattern is consistent with the EDA finding that afternoon deliveries have longer median dwell times.

4.6 Consolidated Feature Importance

Table 4.11 consolidates the feature importance evidence from all three methods; Weibull AFT acceleration factors, RSF VIMP, and XGBoost-AFT SHAP; into a single consensus ranking. Rank 1 = most important. The Weibull rank uses $|\exp(\hat{\beta}) - 1|$ as the effect size measure. The agreement across all three methods provides robust evidence that the ranking is not model-dependent.

Table 4.11: Feature importance consensus across three methods

Covariate / Ranks ->	Weibull	RSF VIMP	SHAP	Consensus
<code>delivery_dayofweek</code>	1	1	1	1
<code>compartment_size_enc</code>	3	2	2	2
<code>delivery_hour</code>	5	3	3	3
<code>piece_size_enc</code>	4	4	4	4
<code>size_match</code>	2	5	5	5
<code>booking_to_delivery_hours</code>	7	6	6	6
<code>delivery_month</code>	6	7	7	7

The consensus ranking is entirely stable across all three methods. This degree of agreement between a parametric AFT model, a non-parametric ensemble, and a gradient-boosted tree model with SHAP attribution; provides strong evidence that the ranking reflects genuine signal in the data rather than artefacts of any single model's assumptions.

The ranking directly answers RQ2. The dominant factor is **delivery day of week**, driven by the Friday effect. The second and third factors are **compartment size** and **delivery hour**, both of which are observable at the time of delivery and therefore actionable for operational decisions. **Piece size** ranks fourth. The bottom two features; booking-to-delivery lead time and delivery month; contribute negligible predictive value across all methods and can be safely excluded from any operational dwell time model without loss of accuracy.

The stability of this ranking also has a methodological implication: the four main predictors are all temporal or size-related features, consistent with the hypothesis that pickup behaviour is primarily governed by *when* in the customer's routine the parcel arrives and *how much effort* collection requires. Features that are not in the control; customer schedule, distance, personal urgency; remain unobserved and constitute the dominant source of unexplained variance that limits the C-index ceiling to approximately 0.66 regardless of model complexity.

4.7 Summary and Answers to Research Questions

RQ1: How can the probability distribution of pickup times be modelled? All five covariate models produce valid survival curve estimates that are substantially better calibrated than the marginal KM baseline (IBS well below 0.25). The Weibull AFT model is the preferred

parametric model on distributional fit grounds ($\Delta\text{AIC} = 8,751$ over Log-Normal). XGBoost-AFT achieves the best overall calibration ($\text{IBS} = 0.1498$) and the best discrimination ($\text{C-index} = 0.6627$). For segment-level modelling, the stratified KM analysis demonstrates that Friday deliveries and large parcels have distinctly different pickup time distributions that can be characterised accurately with simple non-parametric estimators.

RQ2: What factors most influence dwell time? Three sources of evidence converge on the same answer: the Weibull acceleration factors, the RSF VIMP scores, and the XGBoost-AFT SHAP values all rank `delivery_dayofweek` as the dominant predictor, followed by `compartment_size_enc`, `delivery_hour`, and `piece_size_enc`. The Friday delivery effect accounts for the largest single source of variation in dwell time, with Friday parcels having median dwell times 1.5 times longer than Monday parcels. Booking-to-delivery lead time and delivery month carry no meaningful predictive information in this dataset.



5 Conclusion

This thesis aimed to develop and evaluate statistical and machine learning models for predicting parcel dwell time in PostNord’s locker network, and to identify which features most reliably predict fast versus slow customer pickup. Two research questions structured the work, and this chapter summarises the answers to both, the contributions of the thesis, and its principal limitations.

5.1 Answers to the Research Questions

RQ1: How can the probability distribution of pickup times be modelled? Five survival models were implemented and compared. All produce survival curve estimates that are substantially better calibrated than a marginal baseline, with integrated Brier scores of 0.1498 and 0.1507 for the two machine learning models, well below the null reference of 0.25. The Weibull AFT model is the preferred parametric model on distributional fit grounds, with an AIC advantage of 8,751 over the Log-Normal alternative. XGBoost-AFT achieves the best overall performance, with the highest concordance index (0.6627) and the lowest integrated Brier score (0.1498).

However, a central quantitative finding is that all covariate models converge to concordance index values in the narrow range 0.649 to 0.663 regardless of complexity. The marginal gain from moving from a simple Cox proportional hazards model to a tree gradient boosted ensemble is only 0.013 concordance index points. This convergence indicates that the available features of delivery timing, parcel size, and compartment size saturate predictive performance well below what would be needed for reliable individual-level pickup ordering. The data is inherently low-signal with respect to individual customer behaviour.

Despite this ceiling, the models successfully characterise the population-level pickup time distribution. The stratified Kaplan-Meier analysis shows that distinct, well-separated survival curves emerge by delivery day of week and parcel size, meaning that group-level pickup distributions can be predicted with high confidence even when individual predictions are imprecise.

RQ2: What factors influence the dwell time of parcels? Three independent methods; Weibull AFT acceleration factors, Random Survival Forest permutation importance, and XGBoost-AFT SHAP values; produce an identical consensus feature ranking. Delivery day of week is the dominant predictor by a substantial margin. Friday deliveries have median dwell

times of approximately 39 hours, compared to 16 hours for Monday deliveries, a factor of 2.4 driven by the absence of deliveries over the weekend and the correspondingly longer interval before the customer’s next collection opportunity. Compartment size, delivery hour, and piece size are the next three most important factors. Booking-to-delivery lead time and delivery month carry negligible predictive information and can be excluded from any operational dwell time model without loss of accuracy.

The agreement across three methodologically independent importance measures provides robust evidence that this ranking reflects genuine signal in the data rather than an artefact of any single model’s assumptions. The SHAP beeswarm analysis further reveals that the day-of-week effect operates primarily through a discrete Friday discontinuity rather than a gradual linear trend, and that delivery hour’s influence is concentrated in a contrast between morning and afternoon deliveries rather than a continuous gradient.

5.2 Contributions

This thesis makes three contributions to the literature on parcel locker operations and applied survival analysis.

First, it demonstrates that the survival analysis framework is directly applicable to parcel dwell time prediction and produces well-calibrated probabilistic predictions from standard logistics data features.

Second, it provides the multi-model comparative evaluation of dwell time prediction methods on a parcel locker dataset, covering the full spectrum from non-parametric estimators through parametric AFT models to machine-learning-augmented survival models. The finding that all models saturate at approximately the same discrimination level is itself notable: it establishes the information ceiling of the available feature set and identifies unobserved customer behaviour as the dominant source of unexplained variance.

Third, it produces a robust, cross-validated feature importance ranking using three independent methods. The consensus that delivery day of week is the dominant driver, with the Friday effect accounting for the largest identifiable source of variation in dwell time which can be used by the company without any model implementations.

5.3 Limitations

Several limitations should be considered when interpreting these results.

The dataset spans only three months, which limits the diversity of seasonal patterns captured by `delivery_month`. A longer observation window encompassing the pre-Christmas peak period; when locker demand is highest and dwell time patterns may differ most from the average would provide a more representative evaluation.

Five of seven covariates in the Cox proportional hazards model violate the proportional hazards assumption. While the AFT and machine learning models do not require this assumption, the violation indicates that covariate effects on the pickup hazard are genuinely time-varying. This time-varying structure is not fully modelled by any of the models implemented here.

The dataset contains no customer-level features, such as notification preferences, distance from the service point, or purchase category. These unobserved factors are likely the primary drivers of the individual-level variance that limits the concordance index ceiling to approximately 0.66. Any future deployment of dwell time prediction that can incorporate even proxy variables for customer behaviour; such as whether the customer has previously picked up promptly would be expected to achieve substantially higher discrimination.

5.4 Future Work

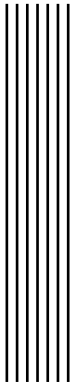
Four concrete directions for future work emerge directly from the findings and limitations above.

Incorporating customer-level features. The most impactful improvement would be to augment the feature set with customer-level information: historical pickup speed for returning customers, distance from the service point estimated from the delivery address, and notification channel or confirmation metadata. Even a single feature capturing whether a customer has previously exhibited fast pickup behaviour would be expected to break the 0.66 concordance index ceiling observed across all models.

Modelling time-varying covariate effects. The Schoenfeld residual test identified time-varying effects for five covariates, most notably `delivery_hour` and `piece_size_enc`. Extended Cox models with time interaction terms, or landmark models that refit the survival function at multiple time horizons, could capture this structure explicitly and would be expected to improve calibration at specific time points, particularly in the first 24 hours where most pickups occur.

Capacity reservation simulation. This thesis produces per-parcel survival curve estimates that are the natural input to a dynamic capacity reservation rule of the form proposed by [6]. A simulation that replays the historical delivery sequence and compares a static first-come first-served assignment strategy against a threshold-based reservation rule derived from the XGBoost-AFT or Weibull AFT survival curves would directly quantify the operational value of the predictive models in terms of acceptance rate and false-acceptance rate.

Deployment on extended real-world data. The current dataset covers three months from a extensive dataset. Retraining on a full calendar year of data, including the pre-Christmas peak period, would allow evaluation of whether seasonal effects suppressed by `delivery_month`'s near-zero importance score in this dataset emerge meaningfully over a longer observation window.



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